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Chip Design

“With Staged **MANUFACTURING PLANS** And **SMT LINES** Being Set Up In India, A Strong **SUPPLY CHAIN** Is Set To Develop Here”

While mid- and high-range smartphones are incorporating intelligent technologies, low-scale phones are also becoming competent. But can we really call these phones intelligent? Neeraj Sharma, country head - India, UNISOC (previously Spreadtrum Communications), talks to Paromik Chakraborty of *Electronics For You* about the current low-price smartphone chip market and the technical projections ahead

Q. How are chipsets evolving in low-range smartphones today?

A. Octa-core is the trend and, hence, chip developers are focussing on that. Their main aim is to bring more performance in chipsets at a price point that was not imaginable earlier. Moreover, they want to ensure that power consumption does not go up with improving chip capabilities.

Q. Is AI integration happening in normal-range smartphones, or is it just a marketing gimmick?

A. Artificial intelligence (AI) is an important trend today. In the new octa-core offerings, developers are incorporating AI algorithms.

It is essential for solution providers to catch up with the newest technologies. However, every new technology needs to have a strong use case before it gets adopted by the mass market. Excluding the tech-savvy generation, most people are not even aware of AI and how it can make their lives easy.

As of now, main applications of AI for smartphones are power efficiency, battery consumption and camera improvement. From a marketing perspective, these are important. However, from end-user perspective, AI should add some extraordinary real-life applications.

Q. How much high-level designing is being done for chipsets in India?

A. India does have a good talent pool, but proportion of development

work for a business depends on the maximum value it can procure from here.

One of our key design teams is based out of Noida, Uttar Pradesh. It designs camera applications. Engineering is done here right from pre-silicon level to the final application stage, including drivers. Even Google Mobile Services applications are done from our India office.

Q. What challenges do chip businesses face in India? How do they address these?

A. We are still many years away from having a full-scale foundry in India. Hence, domestic fab creation will take time. From design point of view, as of now, apart from Samsung, no other mobile manufacturer has a good design team in India.

To address this, partnering with other companies helps. For example, we have a partnership with Lava, in addition to a licence agreement with them. We are collaborating to build a strong design quotient and, in the process, creating smartphones that are completely designed in India.

We may see some phones coming out next year. Right now, the biggest challenge is the learning curve for the both of us, since this is the first time we are developing a complete design within the country.

Going further, with staged manufacturing plans and SMT lines being set up in India, a strong supply chain is set to develop here. I foresee the

supply chain coming by mid-2019, which will give the Indian market a certain price advantage over China.

Some of our supply chain partners who visited Indian factories recently gave me similar feedback. Companies who are setting up production units here will reap benefits soon.

Q. Is there any scope for business in the nearly-saturated smartphone market?

A. It is true that there is a lot of competition, but there is still a lot of scope. In India, there are another 300 to 400 million users who are yet to switch to smartphones.

The focus is currently on making octa-core the prime platform. Even for entry-level smartphones, we are collaborating with partners to put a minimum performance standard in place so that customers have a good experience.

Q. When do you see India getting its own 5G ecosystem?

A. In India, some early deployments may start happening in 2020. Mass-level acceptance for 5G will come by 2022.

Q. Are feature phones dying?

A. Feature phones are still alive. Apart from India, countries in Africa, south-east Asia and Latin America still use feature phones. I think these phones will stay, but 2G phones will not. Most feature phones will become 4G-compatible. **EFY**